

UNIT - II

Fourier Analysis of Continuous Time signals & systems.

Fourier series:

- Periodic signal analysis
- The representation of signal with respect to time is called time domain.
- This Time domain representation is not sufficient for signal analysis. So we take frequency domain.
- The representation of signal with respect to frequency is called frequency domain.
- line spectrum.
- Fourier series: Representation of periodic signal in terms of an infinite sum of sines, cosines & exponential.

Types:

1. Trigonometric FS
2. Exponential FS
3. Cosine FS

Dirichlet condition: for existence of

Fourier series :

Fourier series only if the following Dirichlet conditions are satisfied.

1) signal should be absolutely integrable over the range of time period.

$$\int_0^T |x(t)| dt < \infty$$

2) signal should have finite no of ~~max~~ maxima & minima over the range of time period

3) signal should be have finite no of discontinuities over the range of time period.

Trigonometric Fourier Series :

Representation of Trigonometric FS :

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n \omega_0 t + \sum_{n=1}^{\infty} b_n \sin n \omega_0 t$$

$$a_0 = \frac{1}{T_0} \int_{t_0}^{t_0+T} x(t) dt$$

$$a_n = \frac{2}{T_0} \int_{t_0}^{t_0+T} x(t) \cdot \cos n \omega_0 t dt$$

$$a_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \cdot \cos n \omega_0 t \, dt$$

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \cdot \sin n \omega_0 t \, dt$$

$$\omega_0 = \frac{2\pi}{T} \rightarrow \text{fundamental frequency}$$

$$\rightarrow \text{fundamental Time period.}$$

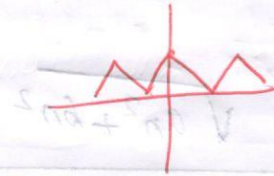
For even signal:

$$x(t) = x(-t)$$

$$a_0 = \frac{2}{T} \int_0^{T/2} x(t) dt$$

$$a_n = \frac{4}{T} \int_0^{T/2} x(t) \cos n\omega_0 t dt$$

$$\boxed{b_n = 0}$$



For odd signal:

$$x(t) = -x(-t)$$

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = \frac{4}{T} \int_0^{T/2} x(t) \sin n\omega_0 t dt$$



Exponential FS:

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$C_n = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jn\omega_0 t} dt$$

$$\omega_0 = \frac{2\pi}{T}$$

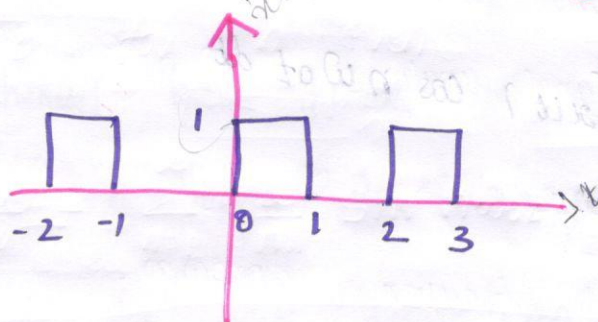
$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi)$$

$$A_0 = \sqrt{a_n^2 + b_n^2}$$

$$\phi = -\tan^{-1}\left(\frac{b_n}{a_n}\right)$$

$$A_n = \sqrt{a_n^2 + b_n^2}$$

1) Find trigonometric FS for the given signal



The representation of trigonometric FS :

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$$

where

$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) dt$$

$$a_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \sin n\omega_0 t dt$$

Here,

$$T = 2$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

$$\boxed{\omega_0 = \pi}$$

$$a_0 = \frac{1}{T} \int_0^T 1 \cdot dt$$

$$= \frac{1}{T} [t]_0^T$$

$$= \frac{1}{T} = \frac{1}{2}$$

$$\boxed{a_0 = \frac{1}{2}}$$

$$a_n = \frac{2}{T} \int_0^T \cos n \omega_0 t \, dt$$

$$= \frac{2}{2} \int_0^1 \cos \pi n t \, dt$$

$$= \frac{2}{2} \left[\frac{\sin n \pi t}{n \pi} \right]_0^1$$

$$= \frac{2}{2} \left[\frac{\sin n \pi}{n \pi} - \frac{\sin 0}{n \pi} \right]$$

$$= 0$$

$$\boxed{a_n = 0}$$

$$b_n = \frac{2}{T} \int_0^T \sin n \pi t \, dt$$

$$= \frac{2}{2} \left[-\frac{\cos n \pi t}{n \pi} \right]_0^1$$

$$= \left[-\frac{\cos n \pi}{n \pi} + \frac{\cos 0}{n \pi} \right]$$

$$\sin n \pi = 0$$

$$\sin 0 = 0$$

$$\boxed{\pi/8 = 1}$$

$$= \left[-\frac{\cos n\pi}{n\pi} + \frac{\cos 0}{n\pi} \right]$$

$$= -\frac{1}{n\pi} [\cos n\pi - \cos 0]$$

$$\cos n\pi = (-1)^n$$

$$\cos 0 = 1$$

$$b_n = \frac{-1}{n\pi} [-1 - 1] \quad ; \text{ when } n = \text{odd}$$

$$\frac{-1}{n\pi} [1 - 1] \quad ; \text{ when } n = \text{even}$$

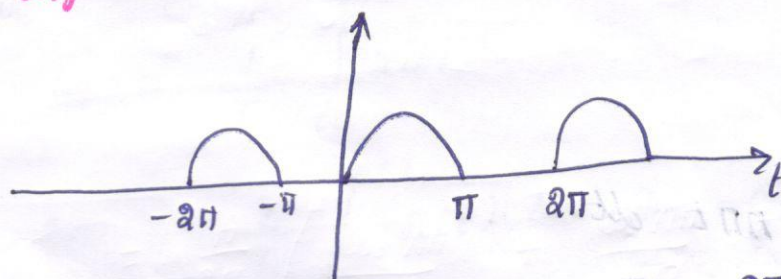
$$b_n = \frac{2}{n\pi} \text{ for } n = \text{odd}$$

$$0 \text{ for } n = \text{even}$$

$$\therefore x(t) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{n\pi} \sin n\omega_0 t \quad [\text{for } n = \text{odd}]$$

$$x(t) = \frac{1}{2} + \frac{2}{\pi} \sin \omega_0 t + \frac{2}{3\pi} \sin 3\omega_0 t + \dots$$

Find trigonometric FS of the half wave rectifier.



$$T = 2\pi$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

For sinusoidal signal, the representation

$$x(t) = A \sin \omega t \quad ; \quad 0 < t < \pi$$

$$0 \quad ; \quad \text{otherwise}$$

The Trig. FS

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$$

i) Find a_0

$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) dt$$

$$= \frac{1}{2\pi} \int_0^{\pi} A \sin \omega_0 t dt$$

$$= \frac{A}{2\pi} \int_0^{\pi} \sin t dt$$

$$= \frac{A}{2\pi} \left[-\cos t \right]_0^{\pi}$$

$$= -\frac{A}{2\pi} [\cos \pi - \cos 0]$$

$$= -\frac{A}{2\pi} [-1 - 1]$$

$$= \frac{-A(-2)}{2\pi} = \frac{A}{\pi}$$

$$\cos \pi = -1$$

$$\cos 0 = 1$$

$$a_0 = \frac{A}{\pi}$$

ii) Find a_n

$$a_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \cos n\omega_0 t dt$$

$$= \frac{2}{2\pi} \int_0^{\pi} A \sin t \cos nt dt$$

Take 2 inside

$$= \frac{A}{2\pi} \int_0^{\pi} 2 \sin t \cos nt \, dt$$

$$\boxed{2 \sin A \cos B = \sin(A+B) + \sin(A-B)}$$

$$= \frac{A}{2\pi} \int_0^{\pi} [\sin(1+n)t + \sin(1-n)t] \, dt$$

$$= \frac{A}{2\pi} \left[-\frac{\cos(1+n)t}{1+n} - \frac{\cos(1-n)t}{1-n} \right]_0^{\pi}$$

$$= \frac{A}{2\pi} \left[-\frac{\cos(1+n)\pi}{1+n} - \frac{\cos(1-n)\pi}{1-n} \right] + \frac{1}{1+n} + \frac{1}{1-n}$$

Multiply the π inside

$$= \frac{A}{2\pi} \left[-\frac{\cos(\pi+n\pi)}{1+n} - \frac{\cos(\pi-n\pi)}{1-n} \right] + \frac{1}{1+n} + \frac{1}{1-n}$$

$$\cos(\pi+n\pi) = \cos n\pi$$

$$\cos(\pi-n\pi) = -\cos n\pi$$

$$= \frac{A}{2\pi} \left[\frac{\cos n\pi}{1+n} + \frac{\cos n\pi}{1-n} + \frac{1}{1+n} + \frac{1}{1-n} \right]$$

If n is even

$$a_n = \frac{A}{2\pi} \left[\frac{2}{1+n} + \frac{2}{1-n} \right]$$

$$= \frac{2A}{2\pi} \left[\frac{1-n+1+n}{1-n^2} \right]$$

$$a_n = \frac{2A}{\pi(1-n^2)} ; n=2, 4, 6, 8$$

If n is odd,

$$\boxed{a_n = 0;}$$

$$\boxed{a_n = \frac{2A}{\pi(1-n^2)} ; n = 2, 4, 6, 8, \dots}$$

Find b_n

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \sin n\omega_0 t \, dt$$

$$= \frac{2}{2\pi} \int_0^{\pi} A \sin t \sin n\omega_0 t \, dt$$

$$= \frac{A}{2\pi} \int_0^{\pi} 2 \sin t \sin nt \, dt$$

$$\boxed{2 \sin A \sin B = \cos(A-B) - \cos(A+B)}$$

$$= \frac{A}{2\pi} \int_0^{\pi} [\cos(1-n)t - \cos(1+n)t] \, dt$$

$$= \frac{A}{2\pi} \left[\frac{\sin(1-n)t}{1-n} - \frac{\sin(1+n)t}{1+n} \right]_0^{\pi}$$

$$= \frac{A}{2\pi} \left[\frac{\sin(1-n)\pi}{1-n} - \frac{\sin(1+n)\pi}{1+n} - \frac{\sin 0}{1-n} + \frac{\sin 0}{1+n} \right]$$

Multiply 'π' inside

$$= \frac{A}{2\pi} \left[\frac{\sin(\pi - n\pi)}{1-n} - \frac{\sin(\pi + n\pi)}{1+n} - \frac{\sin 0}{1-n} + \frac{\sin 0}{1+n} \right]$$

$$\sin (\pi - n\pi) = \sin n\pi$$

$$\sin (\pi + n\pi) = -\sin n\pi$$

$$b_n = \frac{A}{2\pi} \left[\frac{\sin n\pi}{1-n} + \frac{\sin n\pi}{1+n} - \frac{\sin 0}{1-n} + \frac{\sin 0}{1+n} \right]$$

If $n=1$; $b_n = \infty$

For remaining value $b_n = 0$

$$b_1 = \frac{2A}{\pi} \int_0^{\pi} \sin t \, dt$$

so find b_1 value : For that $n=1$

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \sin n\omega t \, dt$$

$\downarrow$
 $n=1$

$$b_1 = \frac{2}{2\pi} \int_0^{\pi} A \sin t \sin t \, dt$$

$$= \frac{A}{2\pi} \int_0^{\pi} 2 \sin t \sin t \, dt = \frac{A}{2\pi} \int_0^{\pi} 2 \sin^2 t \, dt$$

$$= \frac{A}{2\pi} \int_0^{\pi} [1 - \cos 2t] \, dt$$

~~$$2 \sin^2 A = 1 - \cos 2A$$~~

$$2 \sin^2 A = 1 - \cos 2A$$

$$b_1 = \frac{A}{2\pi} \left[t - \frac{\sin 2t}{2} \right]_0^{\pi}$$

$$= \frac{A}{2\pi} \left[\pi - \frac{\sin 2\pi}{2} - 0 + \frac{\sin 0}{2} \right]$$

$$= \frac{A}{2\pi} [\pi]$$

$$\sin \pi = 0$$

$$\sin 0 = 0$$

$$b_1 = \frac{A}{2}$$

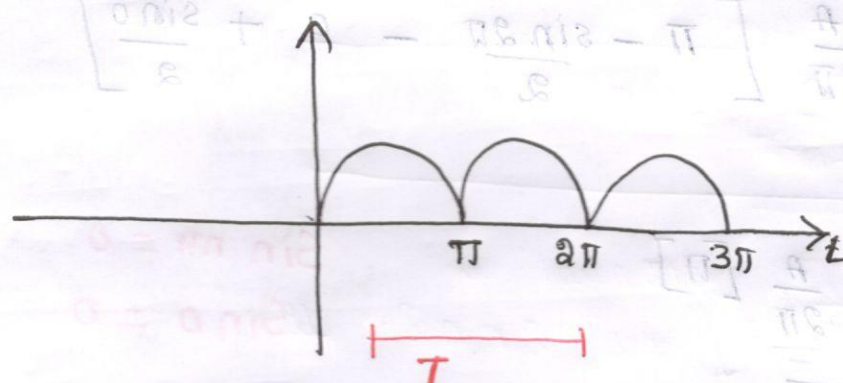
Sub a_0 , a_n , b_1 value in FS trigonometric Representation

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$$

$$x(t) = \frac{A}{\pi} + \frac{A}{2} \sin t + \sum_{n=\text{even}}^{\infty} \frac{2A}{\pi(1-n^2)} \cos nt; \quad 0 < t < 2\pi$$

Even	$x(t) = x(t)$	$a_0 = \frac{2}{T} \int_0^{T/2} x(t) dt$	$a_n = \frac{4}{T} \int_0^{T/2} x(t) \cos n\omega_0 t dt$	$b_n = 0$
odd	$x(t) = -x(-t)$	$a_0 = 0$	$a_n = 0$	$b_n = \frac{4}{T} \int_0^{T/2} x(t) \sin n\omega_0 t dt$

Find Trigonometric FS for the given signal
[Full wave rectifier]



$$T = 2\pi$$

$$\omega_0 = \frac{2\pi}{T} = 1$$

The given signal is an even signal. Because

$$x(t) = x(-t)$$

$$\text{So } \boxed{b_n = 0}$$

Find a_0 :

$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) dt$$

By even property

$$\boxed{a_0 = \frac{2}{T} \int_0^{T/2} x(t) dt}$$

$$a_0 = \frac{2}{2\pi} \int_0^{\pi} A \sin \omega_0 t dt$$

$$= \frac{A}{\pi} \int_0^{\pi} \sin t \, dt$$

$$= \frac{A}{\pi} \left[-\cos t \right]_0^{\pi} = -\frac{A}{\pi} \left[\cos t \right]_0^{\pi}$$

$$\begin{aligned}
 &= -\frac{A}{\pi} [-1-1] \\
 &= \frac{2A}{\pi}
 \end{aligned}$$

$$a_0 = \frac{2A}{\pi}$$

Find a_n :

$$a_n = \frac{2}{T} \int_0^{t+T} A \sin \omega t \cos n \omega t dt$$

By even property

$$a_n = \frac{4}{T} \int_0^{T/2} A \sin \omega t \cos n \omega t dt$$

$$= \frac{4}{2\pi} \int_0^{\pi} A \sin t \cos nt dt$$

Take 2 inside the integral

$$= \frac{2A}{2\pi} \int_0^{\pi} 2 \sin t \cos nt dt$$

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$= \frac{A}{\pi} \int_0^{\pi} [\sin(1+n)t + \sin(1-n)t] dt$$

$$= \frac{A}{\pi} \left[-\frac{\cos(1+n)t}{1+n} - \frac{\cos(1-n)t}{1-n} \right]_0^{\pi}$$

$$= \frac{A}{\pi} \left[\frac{-\cos (1+n)\pi}{1+n} - \frac{\cos (1-n)\pi}{1-n} + \frac{\cos 0}{1+n} \right. \\ \left. + \frac{\cos 0}{1-n} \right]$$

$$= \frac{A}{\pi} \left[\frac{-\cos(\pi+n\pi)}{1+n} - \frac{\cos(\pi-n\pi)}{1-n} + \frac{1}{1+n} + \frac{1}{1-n} \right]$$

$$\begin{aligned} -\cos(\pi+n\pi) &= \cos n\pi \\ \cos(\pi-n\pi) &= -\cos n\pi \end{aligned}$$

$$a_n = \frac{A}{\pi} \left[\frac{\cos n\pi}{1+n} + \frac{\cos n\pi}{1-n} + \frac{1}{1+n} + \frac{1}{1-n} \right]$$

If n is odd,

$$a_n = 0 ; \text{ because } \cos n\pi = (-1)^n =$$

$$\begin{aligned} \text{If } n = \text{odd} \\ a_n &= \frac{A}{\pi} \left[\frac{-1}{1+n} - \frac{1}{1-n} + \frac{1}{1+n} - \frac{1}{1-n} \right] \\ &= 0 \end{aligned}$$

If n is even;

$$a_n = \frac{A}{\pi} \left[\frac{1}{1+n} + \frac{1}{1-n} + \frac{1}{1+n} + \frac{1}{1-n} \right]$$

$$= \frac{A}{\pi} \left[\frac{2}{1+n} + \frac{2}{1-n} \right]$$

$$= \frac{2A}{\pi} \left[\frac{1}{1+n} + \frac{1}{1-n} \right]$$

$$= \frac{2A}{\pi} \left[\frac{1-n+1+n}{1+n^2} \right]$$

$$= \frac{2A}{\pi} \left[\frac{2}{1+n^2} \right]$$

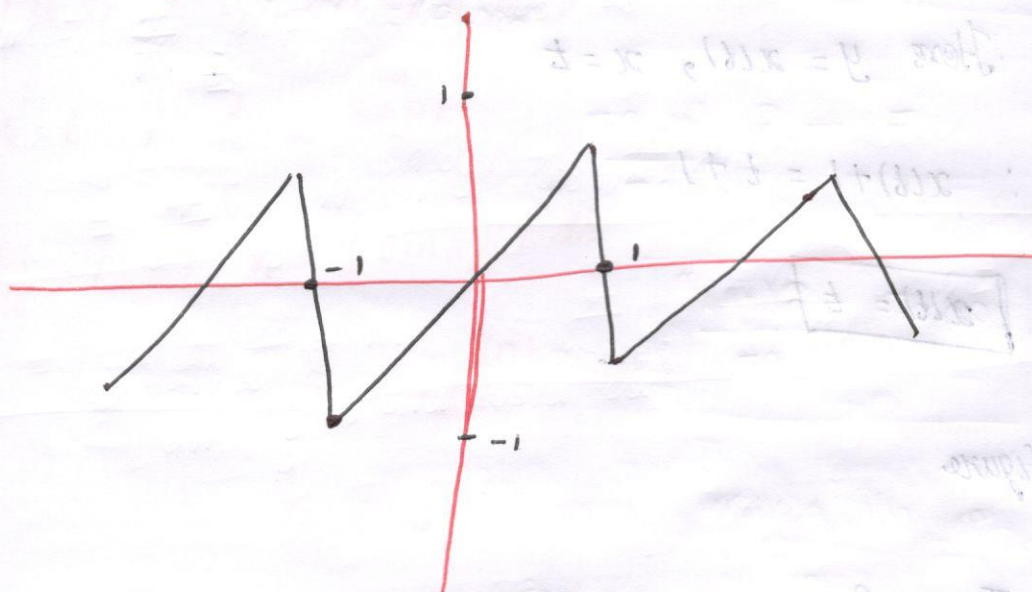
$$A_n = \frac{4A}{\pi(1-n^2)} ; n = 2, 4, 6, \dots$$

The Trigonometry FS for even signal

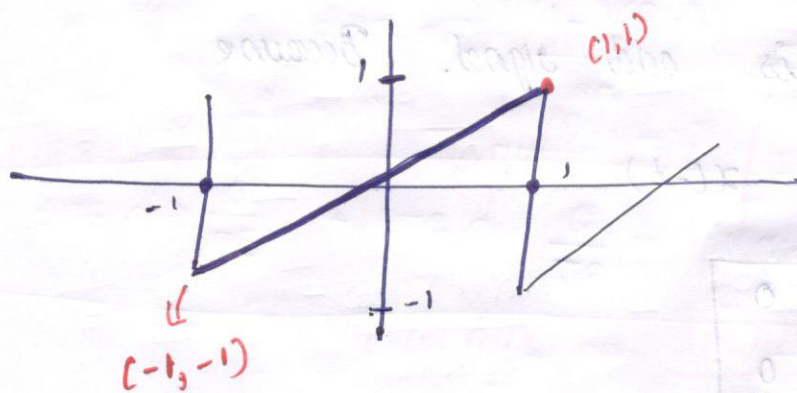
$$x(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos n\omega_0 t$$

$$x(t) = \frac{2A}{\pi} + \sum_{n=\text{even}} \frac{4A}{\pi(1-n^2)}$$

Find the trigonometric FS of the given signal.



From the figure



This is line.

So the line equation

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\frac{y+1}{1+1} = \frac{x+1}{1+1}$$

$$\frac{y+1}{2} = \frac{x+1}{2}$$

$$y+1 = x+1$$

Here $y = x(t)$, $x = t$

$$\therefore x(t)+1 = t+1$$

$$\boxed{x(t) = t}$$

From figure

$$T = 2$$

$$\omega_0 = \frac{2\pi}{T} = \pi$$

The given signal is odd signal. Because

$$x(t) = -x(-t)$$

$$\therefore \boxed{\begin{matrix} a_0 = 0 \\ a_n = 0 \end{matrix}}$$

Find b_n :

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \sin n\omega_0 t \, dt$$
$$= \frac{2}{2} \int_{-1}^1 t \sin n\pi t \, dt$$

$$\int_{-1}^1 t \sin n\pi t \, dt = \int u \, dv$$

$$\int u \, dv = uv_1 - u'v_2 + u''v_3 -$$

$$u = t, \quad dv = \sin n\pi t$$

$$\frac{du}{dt} = 1; \quad v_1 = \int \sin n\pi t \, dt$$

$$= -\frac{\cos n\pi t}{n\pi}$$

$$v_2 = \int v_1$$

$$= -\frac{1}{n\pi} \cdot \frac{\sin n\pi t}{n\pi}$$

$$= -\frac{\sin n\pi t}{(n\pi)^2}$$

$$\int_{-1}^1 t \sin n\pi t \, dt = t \left(-\frac{\cos n\pi t}{n\pi} \right) - \left(-\frac{\sin n\pi t}{(n\pi)^2} \right)$$
$$= -\frac{t \cos n\pi t}{n\pi} + \frac{\sin n\pi t}{(n\pi)^2}$$

$$\therefore b_n = \left[-t \frac{\cos n\pi t}{n\pi} + \frac{\sin n\pi t}{(n\pi)^2} \right]_0^1$$

$$= \left[-\frac{\cos n\pi}{n\pi} + \frac{\sin n\pi}{(n\pi)^2} - \left(\frac{\cos n\pi(-1)}{n\pi} - \frac{\sin n\pi(-1)}{(n\pi)^2} \right) \right]$$

$$= -\frac{\cos n\pi}{n\pi} + 0 - \frac{\cos n\pi}{n\pi} = 0$$

$$\cos(-\theta) = \cos \theta$$

$$= -2 \frac{\cos n\pi}{n\pi}$$

$$b_n = -\frac{2(-1)^n}{n\pi}$$

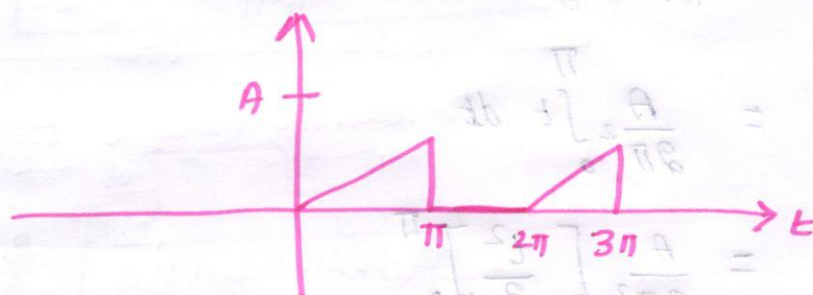
$\therefore$ FS for odd signal

$$x(t) = \sum_{n=1}^{\infty} b_n \sin n\pi t$$

$$x(t) = \sum_{n=1}^{\infty} \frac{(-2)(-1)^n}{n\pi} \sin n\pi t$$

$$x(t) = \frac{2}{\pi} \sin \pi t - \frac{2}{2\pi} \sin 2\pi t + \frac{2}{3\pi} \sin 3\pi t \dots$$

Find trigonometric FS for the below figure.



From the figure,

The points

The line equation is

$$\begin{matrix} (0,0) & (\pi, A) \\ x_1, y_1 & x_2, y_2 \end{matrix}$$

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{y - 0}{A - 0} = \frac{x - 0}{\pi - 0}$$

$$\frac{y}{A} = \frac{x}{\pi}$$

Here $y = x(t)$, $x = t$

$$\frac{x(t)}{A} = \frac{t}{\pi}$$

$$x(t) = \frac{At}{\pi} ; 0 < t < \pi$$

$$T = 2\pi$$

$$\omega_0 = \frac{2\pi}{T} = 1$$

Find a_0

$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) dt$$

$$= \frac{A}{2\pi} \int_0^{\pi} \frac{At}{\pi} dt$$

$$= \frac{A}{2\pi^2} \int_0^{\pi} t dt$$

$$= \frac{A}{2\pi^2} \left[\frac{t^2}{2} \right]_0^{\pi}$$

$$= \frac{A \pi^2}{4\pi^2} = \frac{A}{4}$$

$$\boxed{a_0 = \frac{A}{4}}$$

Find a_n

$$a_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \cos n \omega t dt$$

$$= \frac{2}{2\pi} \int_0^{\pi} \frac{At}{\pi} \cos nt dt$$

$$= \frac{2A}{2\pi^2} \int_0^{\pi} t \cos nt dt$$

$$= \frac{A}{\pi^2} \left[\frac{t \sin nt}{n} + \frac{\cos nt}{n^2} \right]_0^{\pi}$$

$$= \frac{A}{\pi^2} \left[0 + \frac{\cos n\pi}{n^2} - 0 + \frac{1}{n^2} \right]$$

$$a_n = \frac{A}{\pi^2 n^2} [\cos n\pi - 1]$$

If n is even ; $a_n = 0$

If n is odd ; $a_n = \frac{A}{\pi^2 n^2} (-1-1)$

2b (3/17) $\frac{1}{1} = 0.0$

$$a_n = -\frac{2A}{\pi^2 n^2} ; n = \text{odd}$$

Find b_n :

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \sin n\omega_0 t \, dt$$

$$= \frac{2}{2\pi} \int_0^{\pi} \frac{At}{\pi} \sin nt \, dt$$

$$= \frac{A}{\pi^2} \int_0^{\pi} t \sin nt \, dt$$

$$= \frac{A}{\pi^2} \left[-t \frac{\cos nt}{n} - (1) \cdot \frac{1}{n} \frac{\sin nt}{n} \right]_0^{\pi}$$

$$= \frac{A}{\pi^2} \left[-\pi \frac{\cos n\pi}{n} - \frac{1}{n^2} \sin n\pi + 0 \right]$$

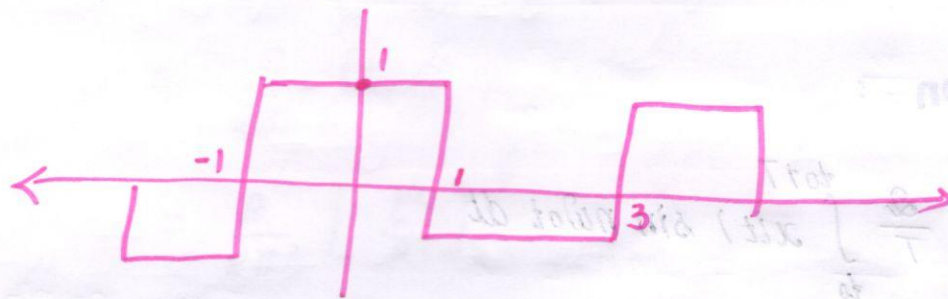
$$= \frac{A}{\pi^2} \left[-\pi \frac{\cos n\pi}{n} \right]$$

$$b_n = -\frac{A \cos n\pi}{n\pi}$$

The trigonometric FS

$$x(t) = \frac{A}{4} + \sum_{n=\text{odd}} \left(-\frac{2A}{n^2\pi^2} \cos nt + \sum_{n=1}^{\infty} \left(-\frac{A \cos n\pi}{n\pi} \right) \sin nt \right)$$

2) Find trigonometric FS



From the figure

The given signal is pulse

$$x(t) = \begin{cases} 1 & \text{for } -1 \leq t \leq 1 \\ -1 & \text{for } 1 \leq t \leq 3 \end{cases}$$

$$\therefore T = 4$$

$$\omega_0 = \frac{2\pi}{T} = \pi/2$$

The given signal is even symmetry.

$$\text{so } b_n = 0$$

Find a_0

$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) dt$$

$$= \frac{1}{4} \int_{-1}^3 x(t) dt$$

$$= \frac{1}{4} \left[\int_{-1}^1 1 \cdot dt + \int_1^3 (-1) dt \right]$$

$$z = \frac{1}{4} \left[[t]_1' + [-t]_1^3 \right]$$

$$= \frac{1}{4} [1 - (-1) + (-3) + 1]$$

$$\boxed{a_0 = 0}$$

$$a_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \cos n \omega_0 t \, dt$$

$$= \frac{2}{4} \int_{-1}^3 x(t) \cos n \omega_0 t \, dt$$

$$= \frac{1}{2} \left[\int_{-1}^1 \cos\left(\frac{n\pi}{2}t\right) dt + \int_1^3 (-1) \cos\left(\frac{n\pi}{2}t\right) dt \right]$$

$$= \frac{1}{2} \left[\left[\frac{\sin \frac{n\pi t}{2}}{n\pi/2} \right]_{-1}^1 - \left[\frac{\sin \frac{n\pi t}{2}}{n\pi/2} \right]_1^3 \right]$$

$$= \frac{1}{2} \times \frac{2}{n\pi} \left[\left[\sin \frac{n\pi t}{2} \right]_{-1}^1 - \left[\sin \frac{n\pi t}{2} \right]_1^3 \right]$$

$$= \frac{1}{n\pi} \left[\sin \frac{n\pi}{2} - \frac{\sin n\pi(-1)}{2} - \frac{\sin 3n\pi}{2} + \frac{\sin n\pi}{2} \right]$$

$$\rightarrow \sin(-\theta) = -\sin \theta$$

$$\sin\left(-\frac{n\pi}{2}\right) = -\sin \frac{n\pi}{2}$$

$$= \frac{1}{n\pi} \left[\sin \frac{n\pi}{2} + \sin \frac{n\pi}{2} - \sin \frac{3n\pi}{2} + \sin \frac{n\pi}{2} \right]$$

$$= \frac{1}{n\pi} \left[3 \sin \frac{n\pi}{2} - \sin \frac{3n\pi}{2} \right]$$

For simplification changing this term to $\sin \frac{n\pi}{2}$

$$\text{So } \sin \frac{3n\pi}{2} = \sin \left(2n\pi - \frac{n\pi}{2} \right)$$

$$= \frac{1}{n\pi} \left\{ 3 \sin \frac{n\pi}{2} - \sin \left(2n\pi - \frac{n\pi}{2} \right) \right\}$$

$$\sin(n\pi \pm \theta) = \pm \sin \theta$$

$$\sin \left(2n\pi - \frac{n\pi}{2} \right) = -\sin \left(\frac{n\pi}{2} \right)$$

$$= \frac{1}{n\pi} \left\{ 3 \sin \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right\}$$

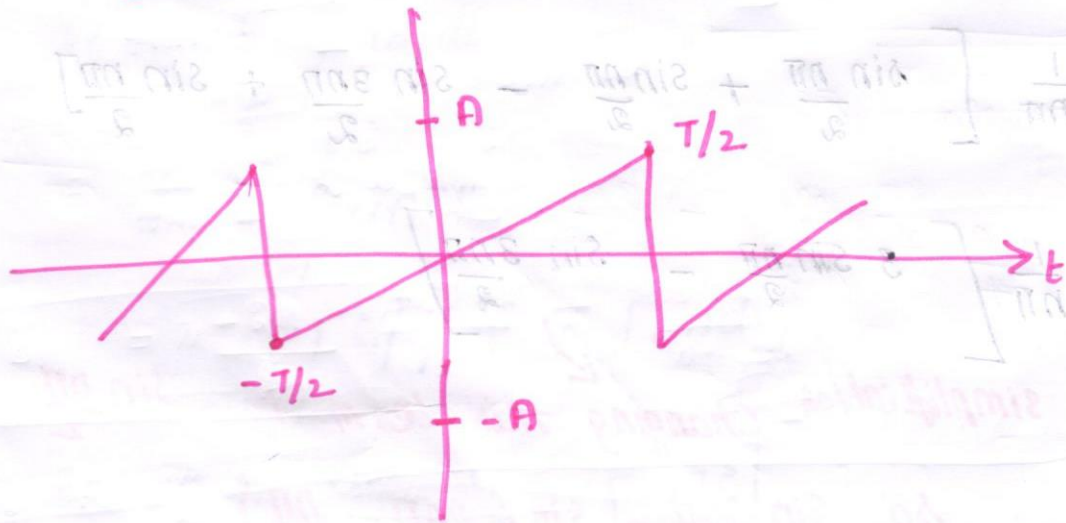
$$= \frac{1}{n\pi} \left[4 \sin \frac{n\pi}{2} \right]$$

$$a_n = \frac{4}{n\pi} \sin \frac{n\pi}{2}$$

$$\therefore x(t) = \sum_{n=1}^{\infty} \frac{4}{n\pi} \sin \left(\frac{n\pi}{2} \right) \cos \left(\frac{n\pi t}{2} \right); -1 < t < 4$$

$$= \frac{4}{\pi} \left[\cos \frac{\pi t}{2} \right]$$

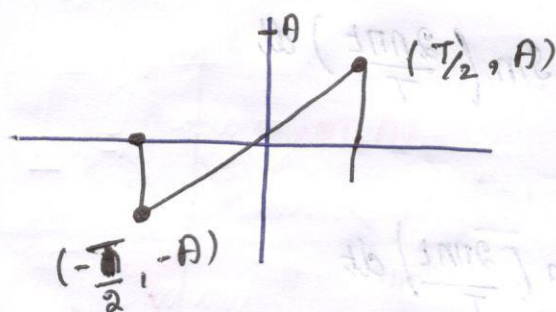
Find trigonometric FS of the signal



From the figure

The given signal is line. The equation for line

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$



$$\begin{matrix} (-\frac{T}{2}, -A) & (\frac{T}{2}, A) \\ x_1 & y_1 & x_2 & y_2 \end{matrix}$$

$$\frac{y + A}{A + A} = \frac{x + T/2}{\frac{T}{2} + T/2}$$

$$\begin{aligned} T &= T \\ \omega_0 &= \frac{2\pi}{T} \end{aligned}$$

$$y = x(t); \quad x = t$$

$$\frac{x(t) + A}{2A} = \frac{t + T/2}{T}$$

$$x(t) + A = \frac{2A}{T} (t + T/2)$$

$$x(t) + A = \frac{2At}{T} + \frac{2AT}{2T}$$

$$x(t) = \frac{2At}{T} + \frac{2AT}{2T} - A$$

$$x(t) = \frac{2At}{T} + A - A$$

$$\boxed{x(t) = \frac{2At}{T} ; \quad -T/2 < t < T/2}$$

The given signal is odd.

$$\text{So } a_0 = 0 ; a_n = 0$$

Find b_n :

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \sin n \omega_0 t \, dt$$

$$= \frac{2}{T} \int_{-T/2}^{T/2} \frac{2At}{T} \sin\left(\frac{2n\pi t}{T}\right) dt$$

$$= \frac{4A}{T^2} \int_{-T/2}^{T/2} t \sin\left(\frac{2n\pi t}{T}\right) dt$$

$$\int u dv = uv_1 - u'v_2 + \dots$$

$$= \frac{4A}{T^2} \left[-t \frac{T}{2n\pi} \cos\left(\frac{2n\pi t}{T}\right) + \frac{T^2}{4n^2\pi^2} \sin\left(\frac{2n\pi t}{T}\right) \right]_{-T/2}^{T/2}$$

$$= \frac{4A}{T^2} \left[\frac{-T^2}{4n\pi} \cos\left(\frac{2n\pi t}{T}\right) + \frac{T^2}{4n^2\pi^2} \sin\left(\frac{2n\pi t}{T}\right) - \frac{T^2}{4n\pi} \cos n\pi \right]$$

$$= \frac{4AT^2}{4n\pi \cdot T^2} \left[-\cos n\pi - \cos n\pi \right]$$

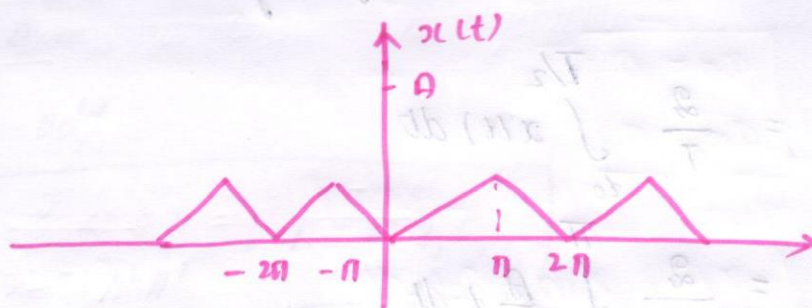
$$= \frac{A}{n\pi} [-2 \cos n\pi]$$

$$b_n = \frac{-2A(-1)^n}{n\pi}$$

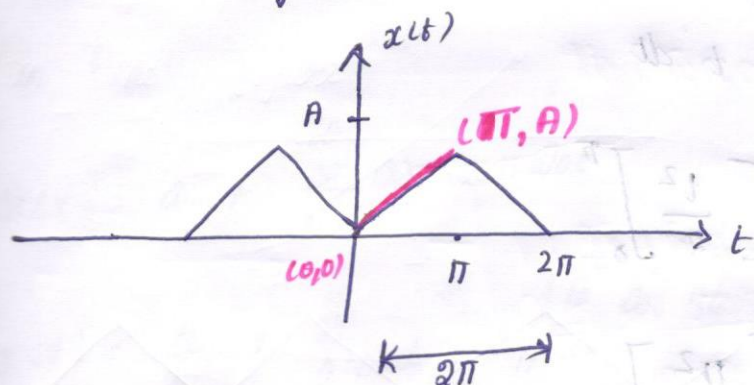
∴ The FS

$$x(t) = \sum_{n=1}^{\infty} -\frac{2A}{n\pi} (-1)^n \sin\left(\frac{2n\pi t}{T}\right); \quad -T/2 < t < T/2$$

Find trigonometric FS of the given signal



From the figure



$$T = 2\pi$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

The given signal is even symmetry. so $b_n = 0$

The given function is line

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1} \quad \begin{matrix} (0,0) & (\pi, A) \\ x_1, y_1 & x_2, y_2 \end{matrix}$$

$$\frac{y - 0}{A - 0} = \frac{x - 0}{\pi - 0}$$

$$\frac{y}{A} = \frac{x}{\pi}$$

$$y = x(t), \quad x = t$$

$$y = x(t), \quad x = t$$

$$\frac{x(t)}{A} = \frac{t}{\pi}$$

$$x(t) = \frac{At}{\pi} ; 0 \leq t < \pi$$

Find a_0

The given signal is even symmetry. So

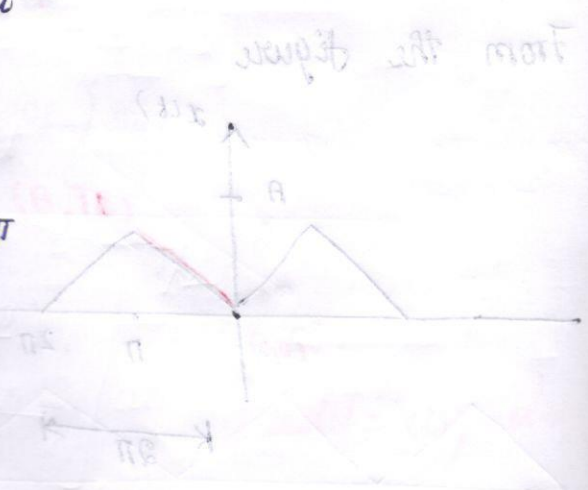
$$a_0 = \frac{2}{T} \int_{t_0}^{T/2} x(t) dt$$

$$= \frac{2}{2\pi} \int_0^{\pi} \frac{A}{\pi} t dt$$

$$= \frac{A}{\pi^2} \int_0^{\pi} t dt$$

$$= \frac{A}{\pi^2} \left[\frac{t^2}{2} \right]_0^{\pi}$$

$$= \frac{A}{\pi^2} \left[\frac{\pi^2}{2} \right]$$



$$a_0 = \frac{A}{2}$$

Find a_n

$$a_n = \frac{4}{T} \int_{t_0}^{T/2} x(t) \cos n\omega t dt$$

$$= \frac{4}{2\pi} \int_0^{\pi} \frac{At}{\pi} \cos nt dt = \frac{4A}{\pi^2} \int_0^{\pi} t \cos nt dt$$

$$= \frac{2A}{\pi^2} \int_0^{\pi} t \cos nt dt$$

u v

$$= \frac{2A}{\pi^2} \left[\frac{t \sin nt}{n} + \frac{\cos nt}{n^2} \right]_0^\pi$$

$$a_n = \frac{2A}{\pi^2} \left[\frac{\cos n\pi}{n^2} - \frac{1}{n^2} \right]$$

If n is even;

$$a_n = 0$$

If n is odd;

$$a_n = -\frac{4A}{n^2\pi^2}; \quad n = 1, 3, 5, \dots$$

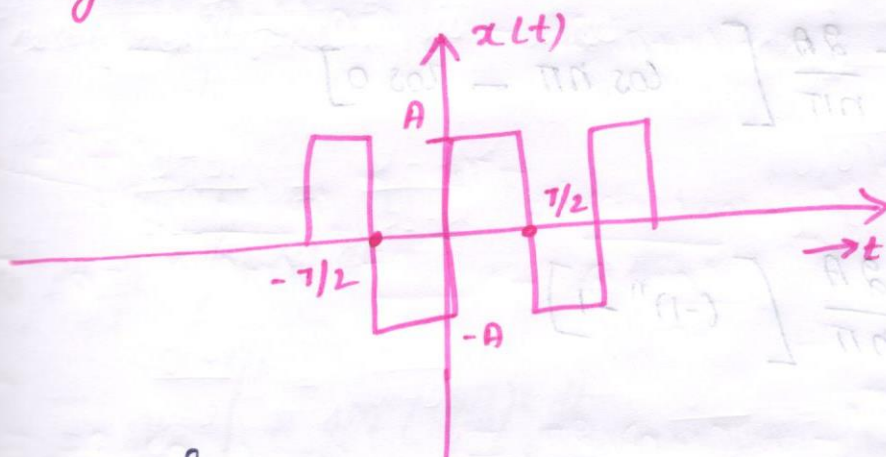
The trigonometric FS

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t$$

$$x(t) = \frac{A}{2} + \sum_{n=\text{odd}}^{\infty} -\frac{4A}{n^2\pi^2} \cos nt; \quad -\pi \leq t < \pi$$

$$x(t) = \frac{A}{2} - \frac{4A}{\pi^2} \left(\cos t - \frac{1}{3^2} \cos 3t + \frac{1}{5^2} \cos 5t + \dots \right)$$

Find Trigonometric FS of the given signal.



$$x(t) = \begin{cases} -A & ; -T/2 < t < T/2 \\ A & ; 0 < t < T/2 \end{cases}$$

$$T = T$$

$$\omega_0 = \frac{2\pi}{T}$$

This is odd symmetry,

$$\boxed{\begin{matrix} a_0 = 0 \\ a_n = 0 \end{matrix}}$$

$$b_n = \frac{4}{T} \int_0^{T/2} x(t) \sin n\omega_0 t \, dt$$

$$= \frac{4}{T} \int_0^{T/2} A \sin n\omega_0 t \, dt$$

$$= \frac{4A}{T} \left[-\frac{\cos n\omega_0 \left(\frac{T}{2} \right)}{\frac{2n\pi}{T}} \right]_0^{T/2}$$

$$= -\frac{4A}{T} \cdot \frac{T}{2n\pi} \left[\cos \frac{2n\pi \cdot T}{2} - \cos 0 \right]$$

$$= -\frac{2A}{n\pi} \left[\cos n\pi - \cos 0 \right]$$

$$b_n = -\frac{2A}{n\pi} \left[(-1)^n - 1 \right]$$

For even

$$b_n = 0$$

For odd

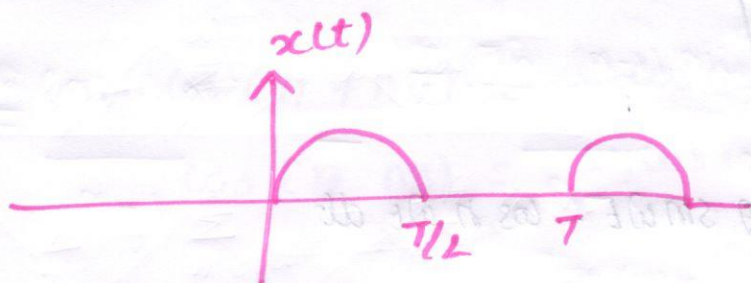
$$b_n = \frac{-2A}{n\pi} [-1 - 1] = \frac{4A}{n\pi}$$

$$b_n = \frac{4A}{n\pi}; \quad n = 1, 3, 5, \dots$$

$$x(t) = \sum_{n=\text{odd}}^{\infty} b_n \sin n\omega t$$

$$x(t) = \sum_{n=\text{odd}}^{\infty} \frac{4A}{n\pi} \sin n\omega t$$

Find trigonometry FS of the given signal



$$x(t) = \begin{cases} A \sin \omega t & 0 < t < T/2 \\ 0 & T/2 < t < T \end{cases}$$

$$T = T$$

$$\omega_0 = \frac{2\pi}{T}$$

Find a_0

$$a_0 = \frac{2}{T} \int_0^T x(t) dt$$

$$= \frac{2}{T} \int_0^{T/2} A \sin \left(\frac{2\pi}{T} t \right) dt$$

$$= \frac{2A}{T} \left[-\frac{\cos \left(\frac{2\pi}{T} t \right)}{\frac{2\pi}{T}} \right]_0^{T/2}$$

$$= \frac{-2A}{T \times \frac{2\pi}{T}} \left[\cos \left(\frac{2\pi}{T} \cdot \frac{T}{2} \right) - \cos 0 \right]$$

$$= \frac{-A}{\pi} [\cos \pi - \cos 0]$$

$$= \frac{-A}{\pi} [-1 - 1] = \frac{-A}{\pi} [-2]$$

$$\boxed{a_0 = \frac{2A}{\pi}}$$

Find a_n

$$a_n = \frac{2}{T} \int_0^{T/2} A \sin \omega t \cdot \cos n \omega t dt$$

$$= \frac{2A}{T} \int_0^{T/2} \sin \omega t \cdot \cos n \omega t dt$$

$$\boxed{2 \sin A \cos B = \sin(A+B) - \sin(A-B)}$$

$$a_n = \frac{A}{T} \left[\int_0^{T/2} \sin(1+n)\omega t - \int_0^{T/2} \sin(1-n)\omega t \right] dt$$

$$= \frac{A}{T} \left[\left[-\frac{\cos(1+n)\omega t}{(n+1)\omega} \right]_0^{T/2} - \left[\frac{\cos(1-n)\omega t}{(1-n)\omega} \right]_0^{T/2} \right]$$

$$= \frac{A}{T \times \omega} \left[\left(-\frac{\cos(1+n) \frac{2\pi}{T} \cdot \frac{T}{2}}{1+n} \right) - \left(\frac{\cos(1-n) \frac{2\pi}{T} \cdot \frac{T}{2}}{1-n} \right) \right]$$

$$+ \left[\frac{1}{1+n} + \frac{1}{1-n} \right]$$

$$= \frac{A}{T \times \frac{2\pi}{T}} \left[\cos \right]$$

$$= \frac{A}{T \times \frac{2\pi}{T}} \left[-\frac{\cos((1+n)\pi)}{1+n} - \frac{\cos((1-n)\pi)}{1-n} + \frac{1}{1+n} + \frac{1}{1-n} \right]$$

$$= \frac{A}{2\pi} \left[\frac{\cos n\pi}{1+n} - \frac{\cos n\pi}{1-n} + \frac{1}{1+n} - \frac{1}{1-n} \right]$$

$$-\cos(\pi + n\pi) = -\cos n\pi$$

$$\cos(\pi - n\pi) = -\cos n\pi$$

If n is odd

$$a_n = 0$$

If n is even

$$a_n = \frac{A}{2\pi} \left[\frac{2}{1+n} + \frac{2}{1-n} \right]$$

$$a_n = \frac{2A}{\pi(1-n^2)}$$

Find b_n :

$$b_n = \frac{2}{T} \int_0^T x(t) \cdot \sin n\omega t \, dt$$

$$= \frac{2}{T} \int_0^{T/2} A \sin \omega t \cdot \sin n\omega t \, dt$$

$$= \frac{A}{T} \int_0^{T/2} 2 \sin \omega t \sin n\omega t \, dt$$

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$= \frac{A}{T} \left[\int_0^{T/2} \cos(1-n)\omega t \, dt - \int_0^{T/2} \cos(1+n)\omega t \, dt \right]$$

$$= \frac{A}{T} \left[\left[\frac{\sin(1-n)\omega t}{(1-n)\omega} \right]_0^{T/2} - \left[\frac{\sin(1+n)\omega t}{(1+n)\omega} \right]_0^{T/2} \right]$$

$$= \frac{A}{T \cdot \omega} \left[\frac{\sin(1-n) \frac{2\pi}{T} \cdot \frac{T}{2}}{1-n} - \frac{\sin(1+n) \frac{2\pi}{T} \cdot \frac{T}{2}}{1+n} + 0 \right]$$

$$= \frac{A}{T \times \frac{2\pi}{T}} \left[\frac{\sin(1-n)\pi}{1-n} - \frac{\sin(1+n)\pi}{1+n} \right]$$

$$= \frac{A}{2\pi} \left[\frac{\sin(\pi - n\pi)}{1-n} - \frac{\sin(\pi + n\pi)}{1+n} \right]$$

$$\sin(\pi - n\pi) = -\sin n\pi$$

$$\sin(\pi + n\pi) = \sin n\pi$$

$$b_n = \frac{A}{2\pi} \left[\frac{\sin n\pi}{1-n} + \frac{\sin n\pi}{1+n} \right]$$

$$\text{if } n=1 ; b_n = \infty$$

For remaining value $b_n = 0$

$$\text{because } \boxed{\sin n\pi = 0}$$

For $n=1$

$$b_n = \frac{2}{T} \int_0^T x(t) \sin \omega t \, dt$$

$$b_1 = \frac{2}{T} \int_0^{T/2} A \sin \omega t \cdot \sin \omega t \, dt$$

$$b_1 = \frac{2A}{T} \int_0^{T/2} \sin^2 \omega t \, dt$$

$$b_1 = \frac{2A}{T} \int_0^{T/2} \left(t - \frac{\cos 2\omega t}{2} \right) dt$$

$$b_1 = \frac{A}{T} \int_0^{T/2} dt - \frac{A}{T} \int_0^{T/2} \cos 2\omega t \, dt$$

$$b_1 = \left[\frac{A}{T} [t]_0^{T/2} - \left[\frac{A}{T} \left(\frac{\sin 2\omega t}{2\omega} \right) \right]_0^{T/2} \right]$$

$$\begin{aligned}
 b_1 &= \frac{A}{T} \left[\frac{T}{2} - 0 \right] - \frac{A}{T} \left[\sin \frac{2\omega t}{2} \right]_0^{T/2} \\
 &= \frac{A}{T} \left[\frac{T}{2} - 0 \right] - \frac{A}{2\omega T} \left[\sin 2 \cdot \frac{2\pi}{T} \cdot \frac{T}{2} - 0 \right] \\
 &= \frac{A}{T} \cdot \frac{T}{2} - \frac{A}{2T\omega} [\sin 2\pi]
 \end{aligned}$$

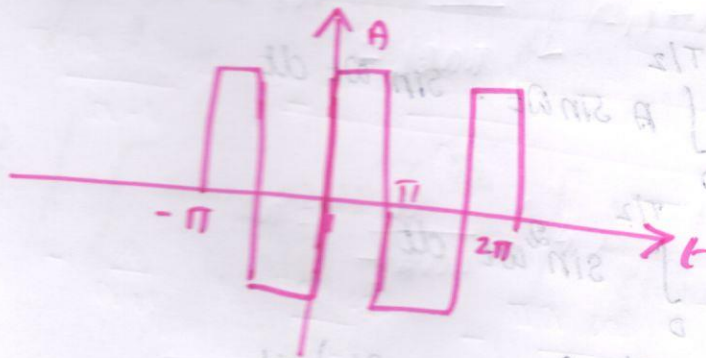
$$b_1 = \frac{A}{2}$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t + b_n \sin n\omega t$$

$$x(t) = \frac{2A}{\pi} + \frac{A}{2} \sin n \left(\frac{2\pi}{T} \right) t + \sum_{n=\text{even}}^{\infty} \frac{2A}{\pi(1-n^2)} \cos \left(\frac{2\pi}{T} \right) t$$

Find exponential
given signal.

Fourier Series for the



The exponential FS

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

$$C_0 = \frac{1}{T} \int_0^T x(t) \cdot dt$$

$$x(t) = \begin{cases} A & 0 < t < \pi \\ -A & \pi < t < 2\pi \end{cases}$$

$$T = 2\pi$$

$$\omega_0 = \frac{2\pi}{T}$$

$$C_0 = \frac{1}{T} \int_0^T x(t) dt$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x(t) dt$$

$$\frac{A}{C} = 1$$

$$= \frac{1}{2\pi} \left[\int_0^{\pi} A dt + \int_{\pi}^{2\pi} -A dt \right]$$

$$= \frac{A}{2\pi} \left[(t)_0^\pi - (t)_\pi^{2\pi} \right]$$

$$= \frac{A}{2\pi} \left[\pi - 0 - 2\pi + \pi \right]$$

$$\boxed{C_0 = 0}$$

$$C_n = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jn\omega_0 t} dt$$

$$= \frac{1}{2\pi} \left[\int_0^\pi A \cdot e^{-jnt} dt + \int_\pi^{2\pi} -A e^{-jnt} dt \right]$$

$$= \frac{A}{2\pi} \left[\left(\frac{e^{-jnt}}{-jn} \right)_0^\pi - \left(\frac{e^{-jnt}}{-jn} \right)_\pi^{2\pi} \right]$$

$$= \frac{A}{-j2\pi n} \left[e^{-jn\pi} - e^0 - e^{-jn2\pi} + e^{-jn\pi} \right]$$

$$= \frac{A}{-j2\pi n} \left[(-1)^n - 1 - 1 + (-1)^n \right]$$

$$e^{-jn\pi} = (-1)^n$$

$$e^0 = 1$$

$$e^{-jn2\pi} = 1$$

$$e^{jn2\pi} = 1$$

$$n = \text{even} ; c_n = 0$$

$$n = \text{odd} ; c_n = \frac{A}{-j2n\pi} [(-1)^{n-1} - 1]$$

$$c_n = \frac{A}{-j2n\pi} (-4)$$

$$c_n = \frac{2A}{jn\pi} ; n = \text{odd}$$

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

$$x(t) = \sum_{n=-\infty}^{\infty} \frac{2A}{jn\pi} e^{jnt} ; \text{ for } n = \text{odd}$$